Lecture 2

- We saw that the general solution of a linear ODE can be expressed in terms of complementary and particular solutions.
  - Complementary solutions form a solution space.
  - For 1st-order ODEs the solution space is 1D and for 2nd-order ODEs they are 2D ...more on this today
- We learnt how to use reduction of order to solve a 2nd-order ODE when one complementary solution is known

Lecture 3

We will:

1. Define linear independence of functions and discuss the importance of linear independence of complementary solutions
2. Learn how to use the Wronskian to determine linear independence of a set of functions
3. Connect linear independence with trial solutions

Existence and uniqueness of solutions

Theorem

Consider $(\mathsf{T}y)(t) = g(t)$ where

$$\mathsf{T} = \frac{\mathrm{d}^2}{\mathrm{d}t^2} + f_1(t)\frac{\mathrm{d}}{\mathrm{d}t} + f_0(t)$$

If f_0, f_1 and g are all continuous over $D = [a, b]$, then over D:

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- ODEs: The general solution to $\mathsf{T}y = g$ is a linear sum of 2 linearly independent complementary solutions and a single particular solution.
  - Solutions space (i.e., all complementary solutions) is a 2-dimensional space.

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No, because $y_1 = \dfrac{1}{2} y_2$

The Wronskian

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  - $\mathrm{wr}[y_1, y_2] = 0$ for some t tells you nothing

Theorem (Abel's differential equation identity)

Let y_1 and y_2 be two solutions of $y'' + f_1 y' + f_0 y = 0$. Then

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If we know one solution, say y_1, we can use Abel's identity and Slide 5 to find the second linearly independent solution, y_2, by solving the ODE

$$y_1 y_2' - y_1' y_2 = \mathrm{wr}[y_1, y_2](t_0)\mathrm{e}^{-\int_{t_0}^{t} f_1(\tau)\mathrm{d}\tau}$$

This is equivalent to using reduction of order.

Exercise 1.7: Study linear independence of solutions to below ODE

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which is non-zero for some t, e.g., $\mathrm{wr}(1) = \mathrm{e}^{-b}$.

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Hence, these two solutions are linearly independent and are a basis for the solution space.

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- Note $f_1 = b \Rightarrow \mathrm{e}^{-\int_{t_0}^{t} f_1(\tau)\mathrm{d}\tau} = \mathrm{e}^{-\int_0^t b\mathrm{d}\tau} = \mathrm{e}^{-bt}$

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- Confirming Abel's identity

Exercise 1.8: Find solutions of the form $\phi = r^\lambda$ for the below ODE

$$\frac{\mathrm{d}^2\phi}{\mathrm{d}r^2} + \frac{1}{r}\frac{\mathrm{d}\phi}{\mathrm{d}r} - \frac{n^2}{r^2}\phi = 0, \qquad r > 0, \qquad n \in \{0, 1, 2, \ldots\}^*$$

*This is the Euler equation from, e.g., electrostatics, which we will see again in Topic 5

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$$\lambda(\lambda-1) + \lambda - n^2 = 0$$

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which has solution $\lambda = \pm n$.

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Hence the general solution is $\phi = C_1 r^n + C_2 r^{-n}$.

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so the general solution for $n = 0$ is $\phi = C_1 + C_2 \log r$

Higher-order (linear) ODEs

Theorem (Solutions of an nth order linear ODE/IVP)

Consider $(\mathsf{T}y)(t) = g(t)$ where

$$\mathsf{T} = \frac{\mathrm{d}^n}{\mathrm{d}t^n} + f_{n-1}(t)\frac{\mathrm{d}^{n-1}}{\mathrm{d}t^{n-1}} + \ldots + f_1(t)\frac{\mathrm{d}}{\mathrm{d}t} + f_0(t)\cdot$$

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Higher-order (linear) ODEs

Theorem (Solutions of an nth order linear ODE/IVP)

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- IVPs: When supplemented with initial values $y(t_0), y'(t_0), \ldots,\ y^{(n-1)}(t_0)$ where $t_0 \in D$, it has a unique solution which is n-times differentiable.

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Given n functions $y_1, y_2, \ldots, y_n$, the Wronskian is given by

$$\mathrm{wr}[y_1, y_2, \ldots, y_n](t) = \det \begin{pmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{pmatrix}$$

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Method of trial solutions (higher order)

Exercise 1.9: Find the general solution to the below ODE by trialing $y = \mathrm{e}^{\lambda t}$.

$$y''' + y'' - 4y' - 4y = 0$$

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Reduction of order also works in higher dimensions but we won't cover that in MAST20030.

Summary of Linear ODEs

1. An nth order linear ODE has n *linearly independent* complementary solutions and a single particular solution. It has a unique solution if n initial values are specified.

2. Check for linear independence using the Wronskian.

3. If one complementary solution for a 2nd order ODE is known, the general solution can be found through reduction of order.